

An abstract network diagram on the left side of the slide, featuring various nodes (circles and squares) connected by lines, representing a complex network topology. The diagram is rendered in a lighter shade of the teal background color.

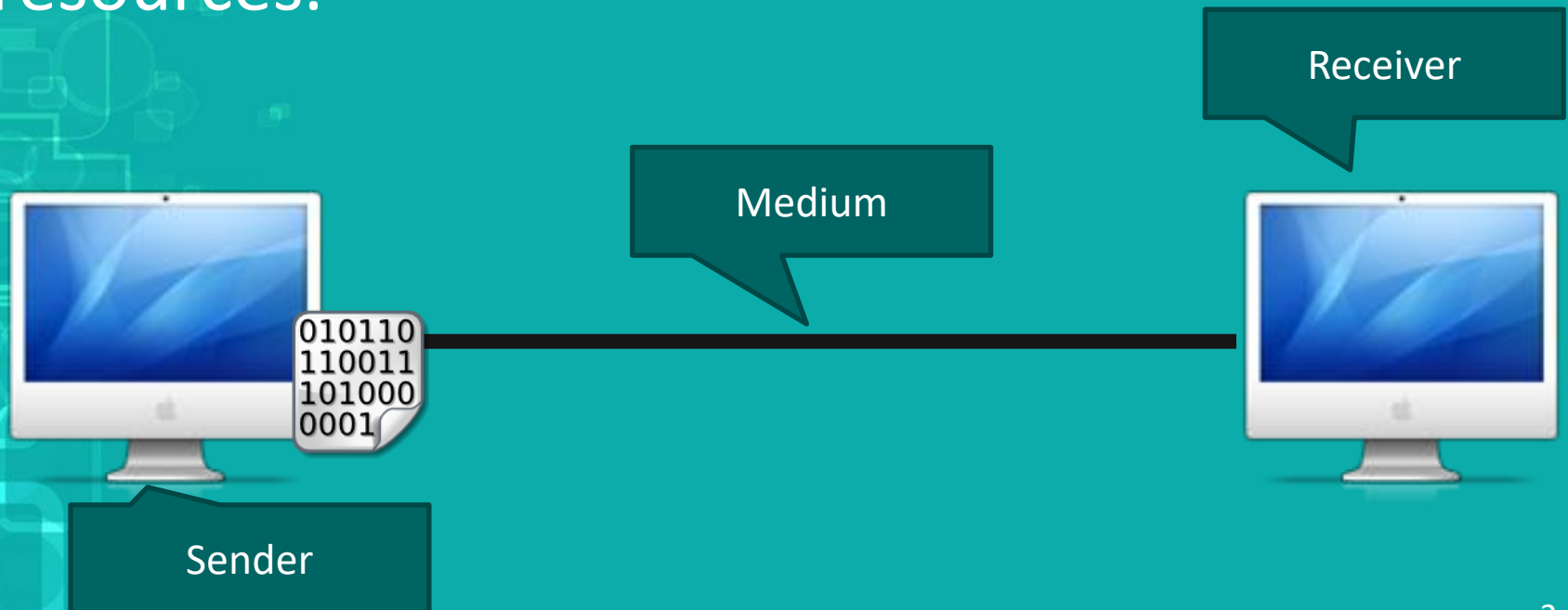
ITM 31023

Computer Networks

Types of Networks & Topology

What is Computer Network?

Network is two or more devices connected to each other with a medium such as wired or wireless. This connection allows those two devices to exchange or share data and resources.



Purposes of having Computer Network

- File sharing
- Printer sharing
- Application services
- Email
- Remote access

What devices make up LANs?

HOSTS



Desktop
PCs



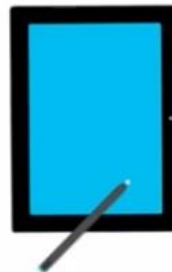
Laptops



Phones



Servers



Tablets

CENTRAL CONNECTING DEVICES



Hub

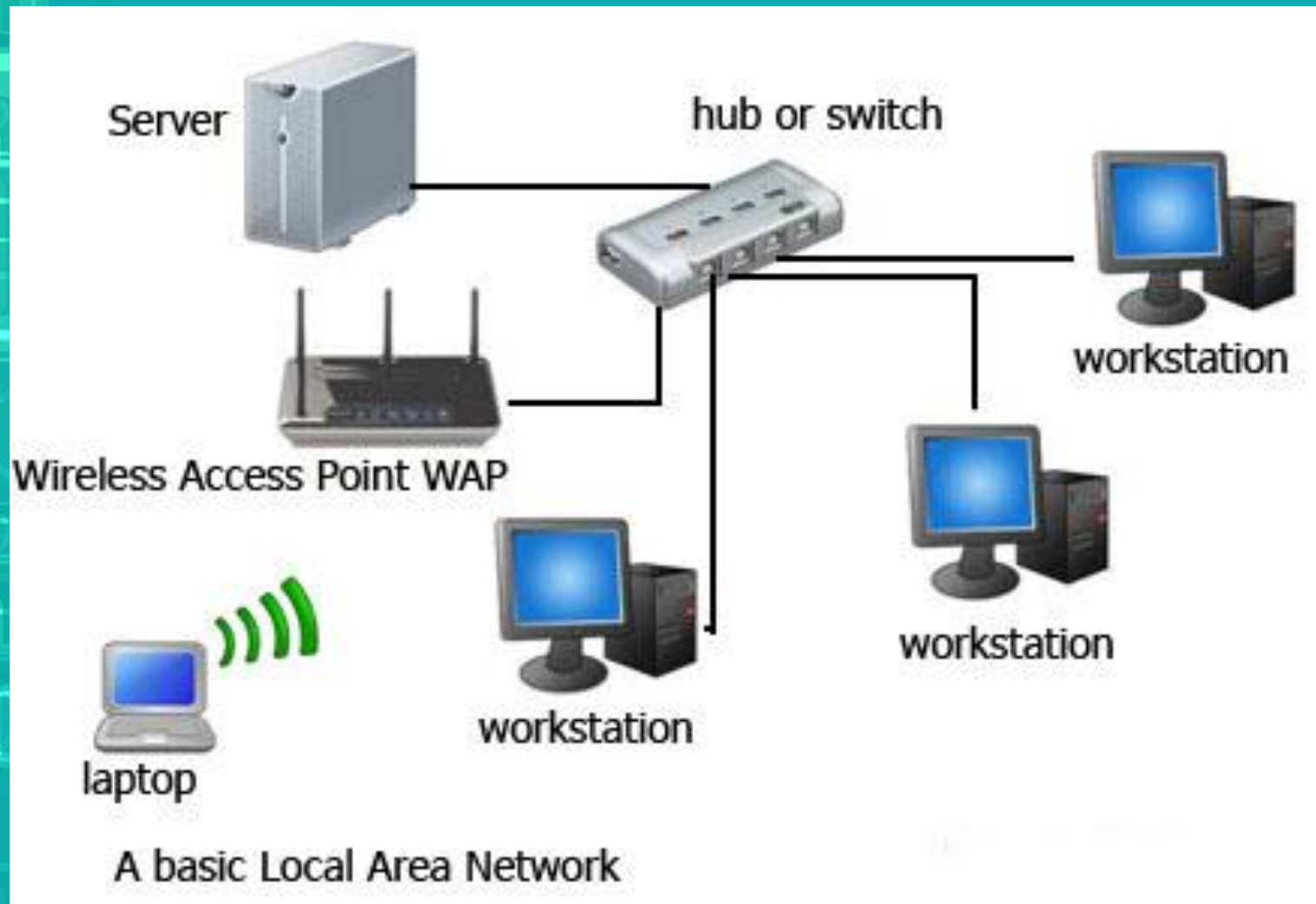


Switch



Router

Simple Local Area Network



How Networks are organized or classified?

Networks are organized or classified based on

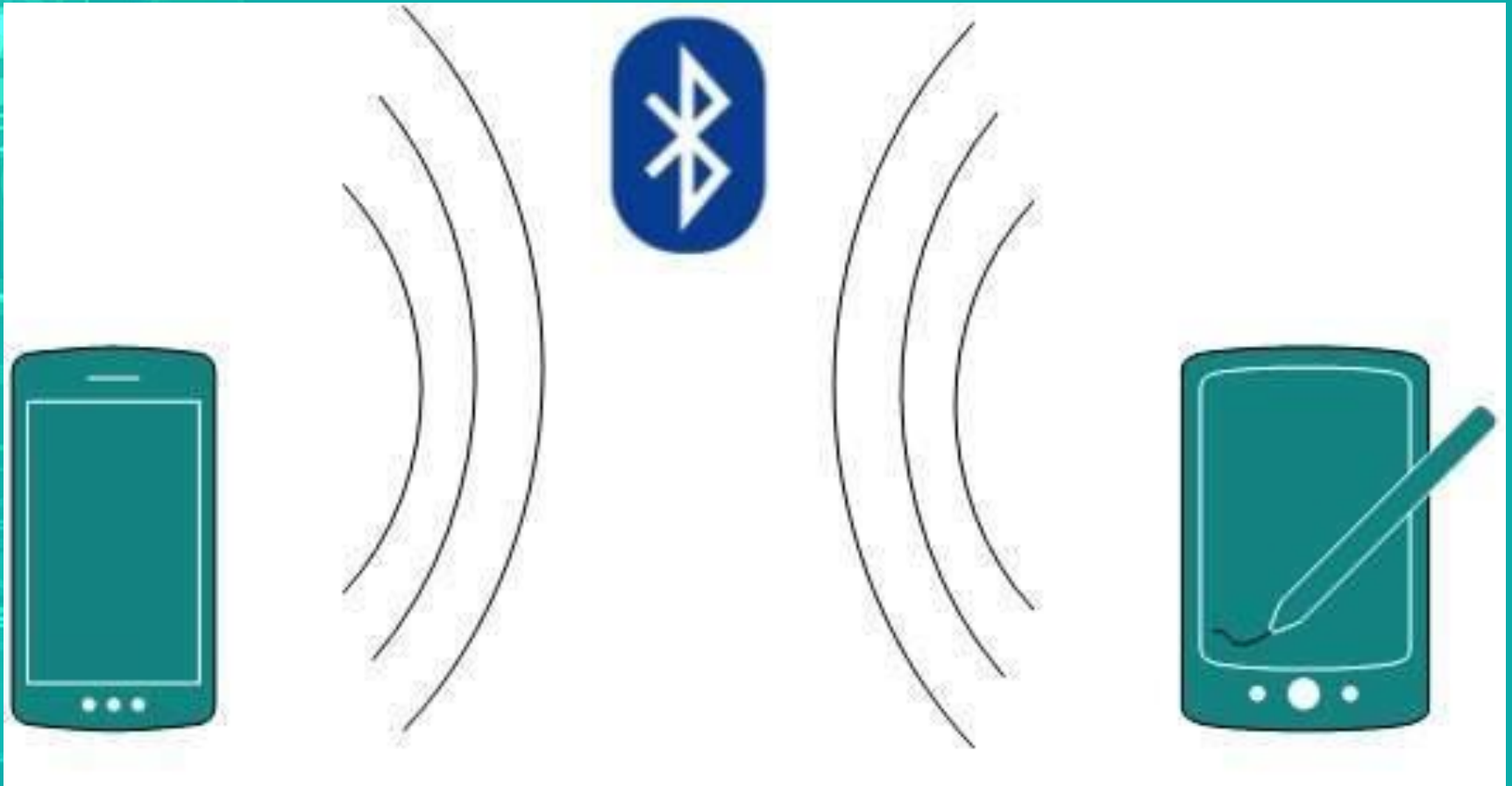
- Geographical span
- Relationship
- Inter-connectivity (Topology)
- Administration

Classifying Network by their Geographical span

The common types are,

- Personal Area Network (PAN)
- **Local Area Network (LAN)**
- **Metropolitan Area Network (MAN)**
- **Wide Area Network (WAN)**

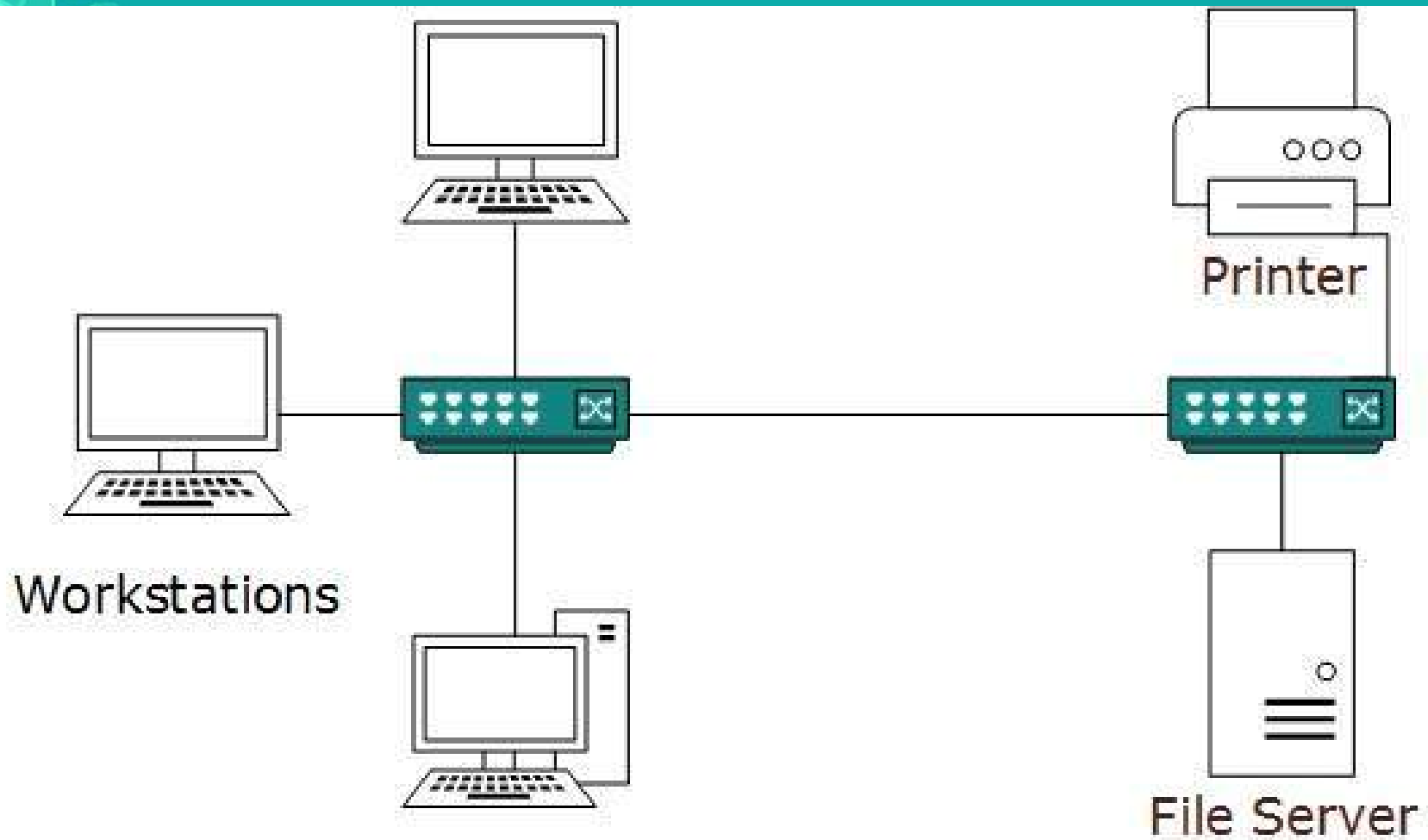
Personal Area Network (PAN)



Personal Area Network (PAN)

A Personal Area Network (PAN) is smallest network which is very personal to a user. This may include Bluetooth enabled devices or infrared enabled devices. PAN may have the connectivity range up to 10 meters. PAN may include wireless computer keyboard and mouse, Bluetooth enabled headphones, wireless printers and TV remotes.

Local Area Network (LAN)



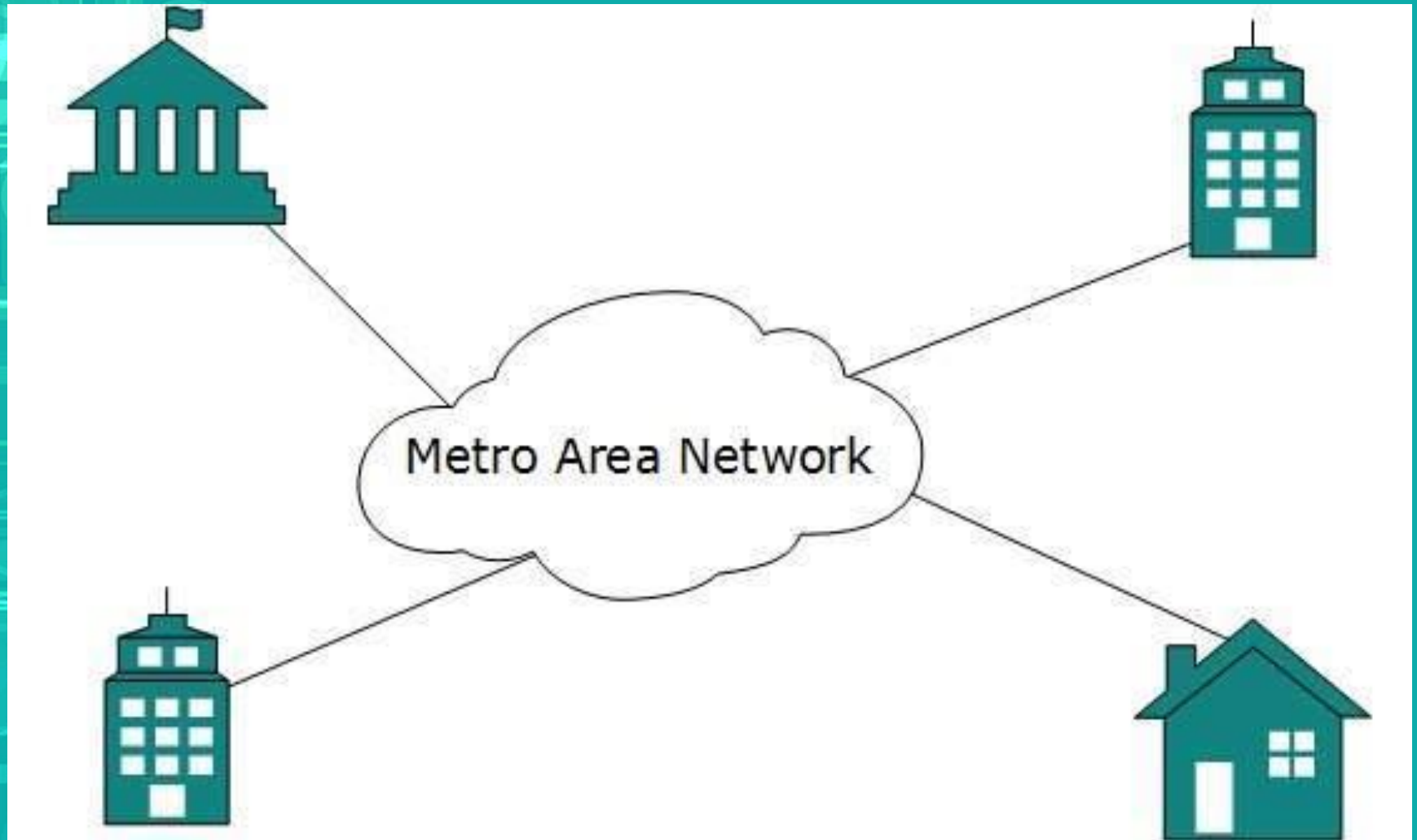
Local Area Network (LAN)

A Local Area Network (LAN) is a computer network that interconnects computers within a limited area such as a residence, school, laboratory, university campus or office building and has its network equipment and interconnects locally managed.

Local Area Network (LAN) contd...

LAN provides a useful way of sharing the resources between end users. The resources such as printers, file servers, scanners, and internet are easily sharable among computers.

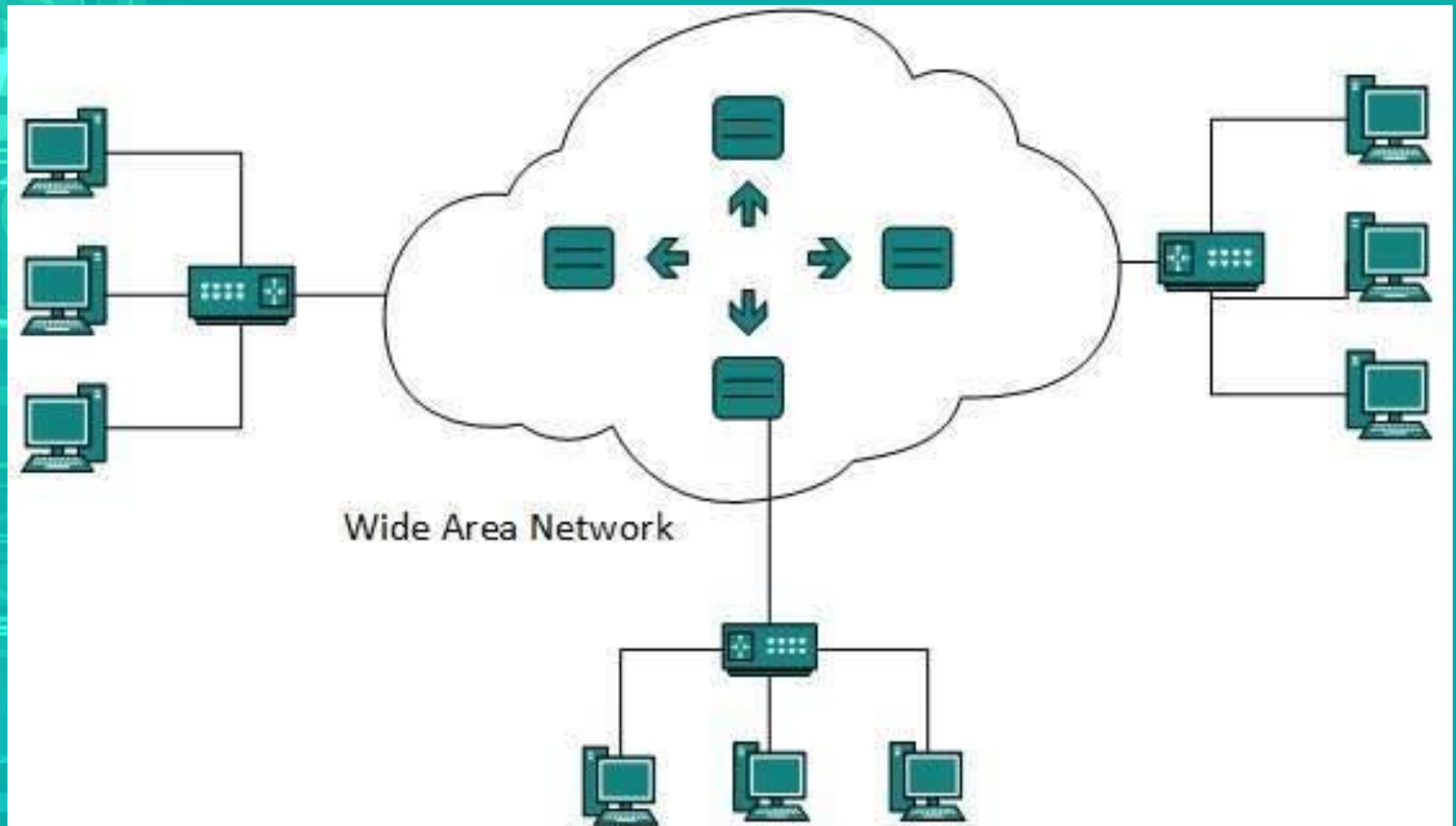
Metropolitan Area Network (MAN)



Metropolitan Area Network (MAN)

- A metropolitan area network, or MAN, consists of a computer network across an entire city, college campus or small region.
- A MAN is often used to connect several LANs together to form a bigger network.
- Depending on the configuration, this type of network can cover an area from several miles to tens of miles

Wide Area Network (WAN)



Wide Area Network (WAN)

- Wide Area Network (WAN) covers a wide area which may span across provinces and even a whole country or continent.
- Generally, telecommunication networks are Wide Area Network.
- These networks provide connectivity to MANs and LANs.
- Since they are equipped with very high speed backbone, WANs use very expensive network equipment.

Classifying Network by their Geographical span - Example

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Network Relationship

Network relationship refers to how one computer makes use of another computer's resources over the network.

Two fundamental types of network relationships exist:

1. Peer-to-peer
2. Client/Server

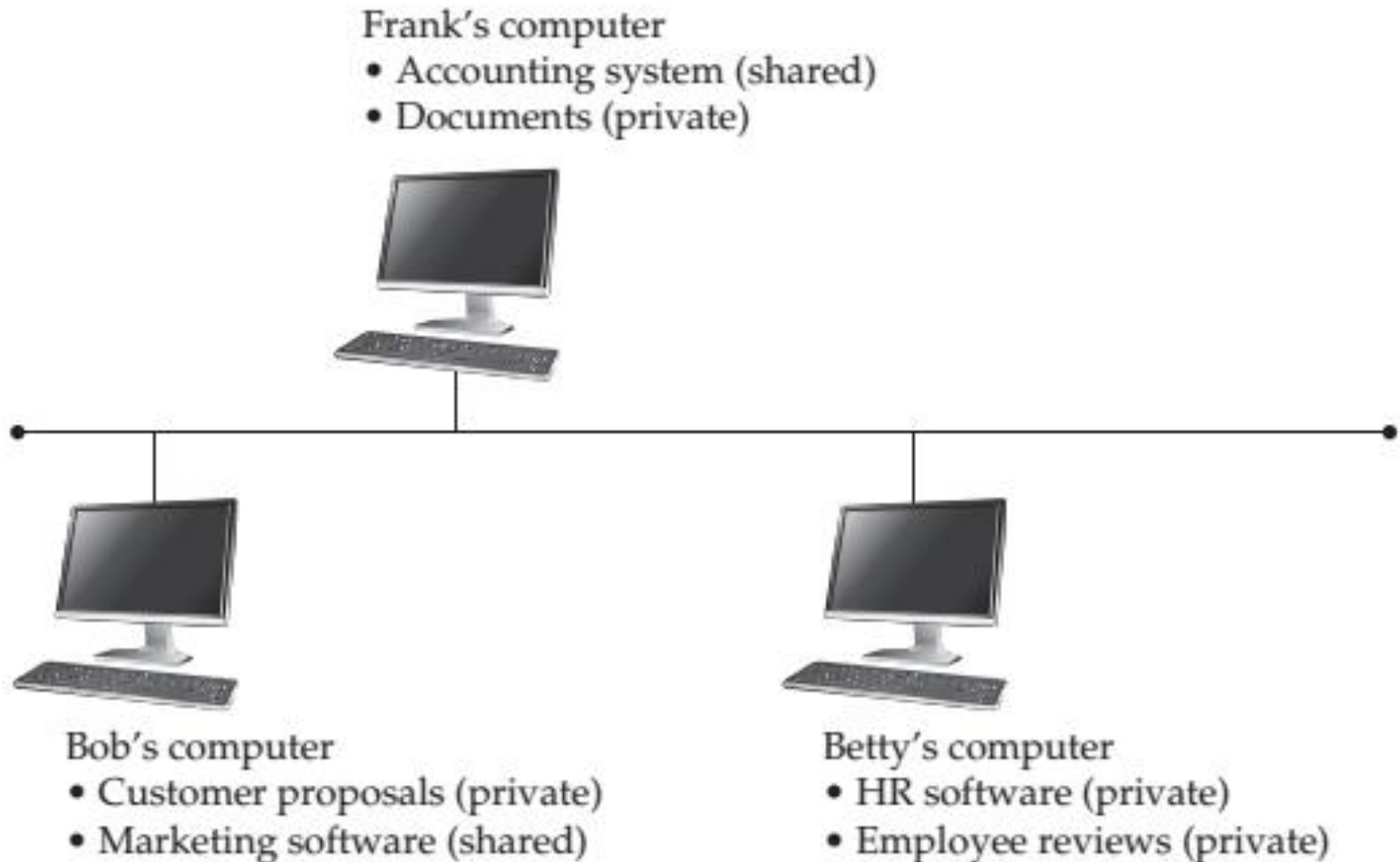
Peer-to-peer

- In a peer-to-peer network relationship, the computers on the network communicate with each other as equals.
- Each computer is responsible for making its own resources available to other computers on the network. These resources might be files, directories, application programs, devices (such as printers, modems, or fax cards), or any combination of these items.

Peer-to-peer

- Each computer is also responsible for setting up and maintaining its own security for those resources.
- Each computer is responsible for accessing the network resources it needs from other peer-to-peer computers, knowing where those resources are located in the network, and handling the security required to access them.

Peer-to-peer



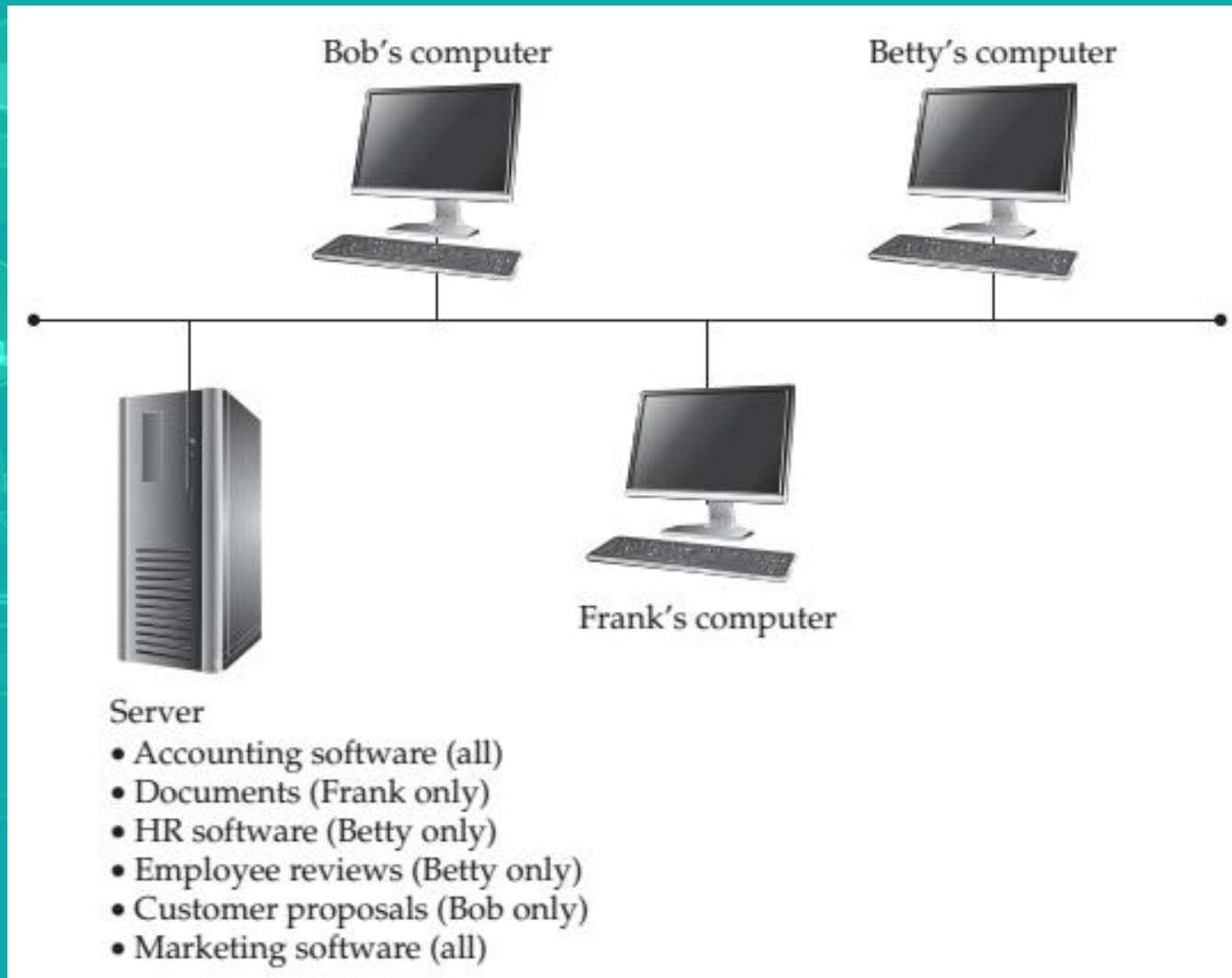
Client/Server

- In a client/server network relationship, a distinction exists between the computers that make network resources available (the servers) and the computers that use the resources (the clients, or workstations).
- The server computers in a client/server network are responsible for making available and managing appropriate shared resources, and for administering the security of those resources.

Client/Server

- A pure client/server network is one in which all available network resources (such as files, directories, applications, and shared devices) are centrally managed and hosted. Those resources are accessed by the client computers.
- None of the client computers share their resources with other client computers or with the servers. The client computers are pure consumers of these shared network resources.

Client/Server



Pros/Advantages of Peer-to-Peer Networks

- Use less expensive computer hardware
- Easy to administer
- No network operating system required
- More built-in redundancy

Cons/Disadvantages of Peer-to-Peer Networks

- Might impact user's performance
- Not very secure
- Difficult to back up
- Hard to maintain version control

Pros/Advantages of Client/Server Networks

- Very secure
- Better performance
- Centralized backup
- Very reliable

Cons/Disadvantages of Client/Server Networks

- Require professional administration
- More hardware-intensive

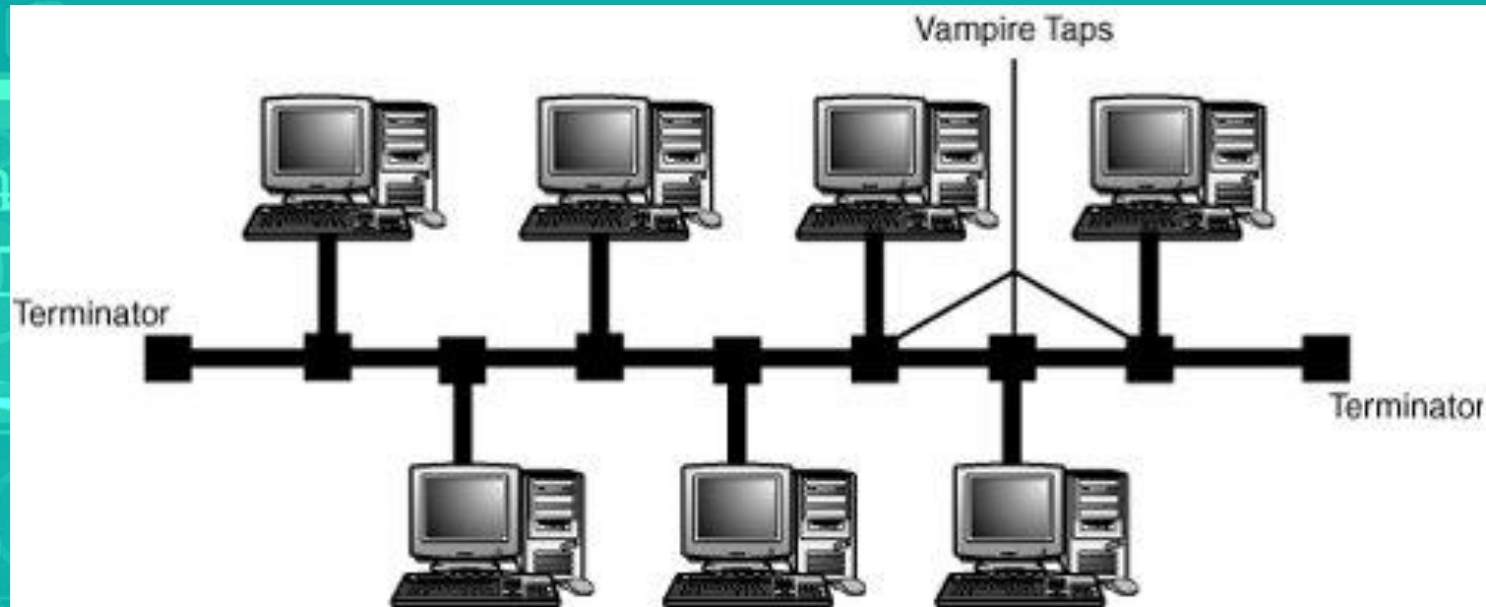
Network Topologies

- Network topology is defines the physical (or logical) connections of hosts in a Network.
- There are multiple types of Network Topologies. Those are:
 - Bus
 - Star
 - Ring
 - Mesh
 - Hybrid

Physical vs. Logical Topology

- The actual layout of a network and its media is its Physical Topology
- The way in which the data access the medium and transmits packets is the Logical Topology
- A glance at a network is not always revealing. Cables emerging from a hub or switch does not make it necessarily a Star Topology – it may actually be a bus or a ring

Bus Topology



Each node is daisy-chained (connected one right after the other) along the same backbone. Information sent from a node travels along the backbone until it reaches its destination.

Bus Topology

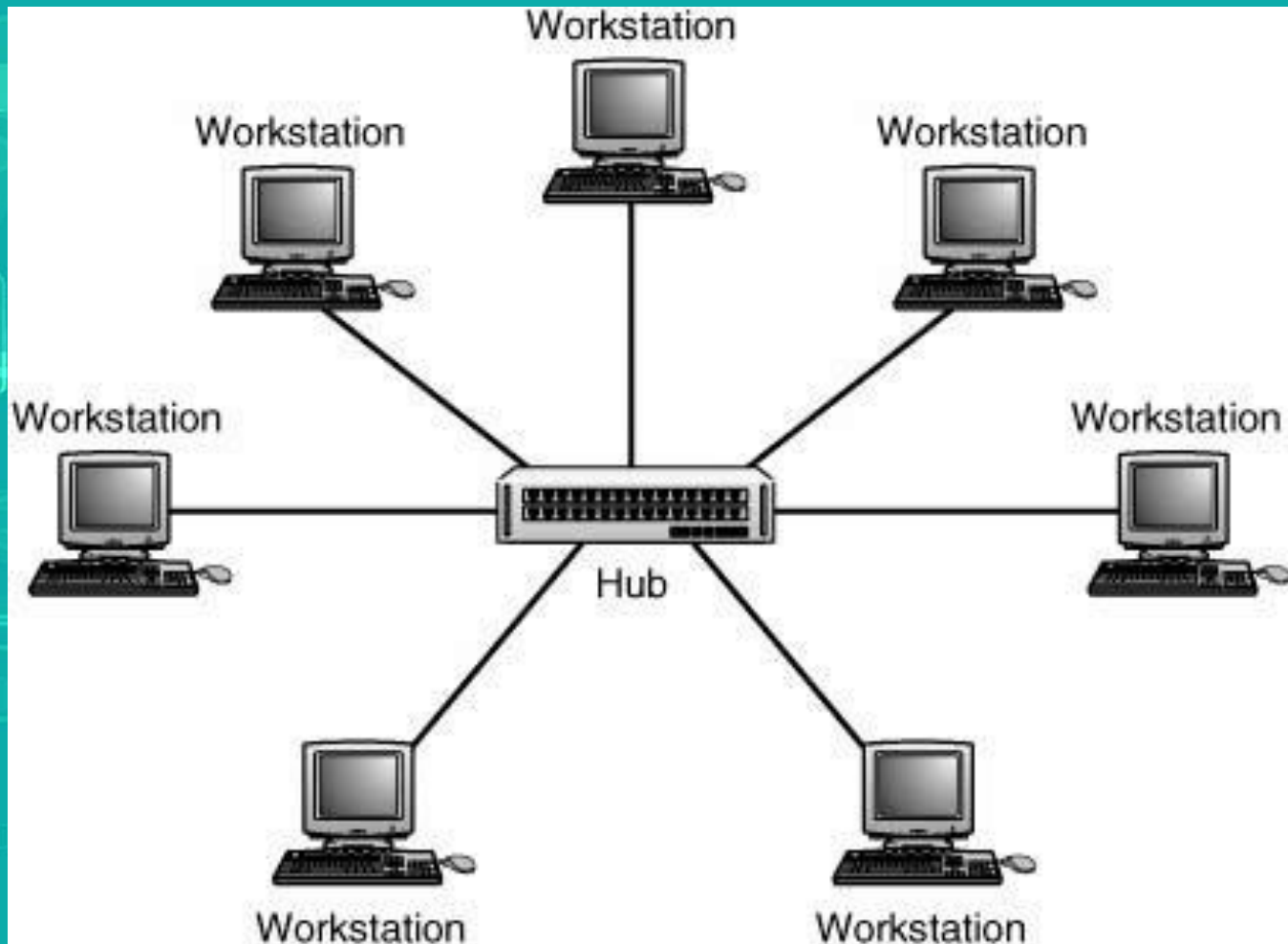
Advantages

- Inexpensive to install
- Easy to add stations
- Use less cable than other topologies
- Works well for small networks

Disadvantages

- Backbone breaks, whole network down
- Limited no of devices can be attached
- Difficult to isolate problems
- Sharing same cable slows response rates
- No longer recommended

Star Topology



Star Topology

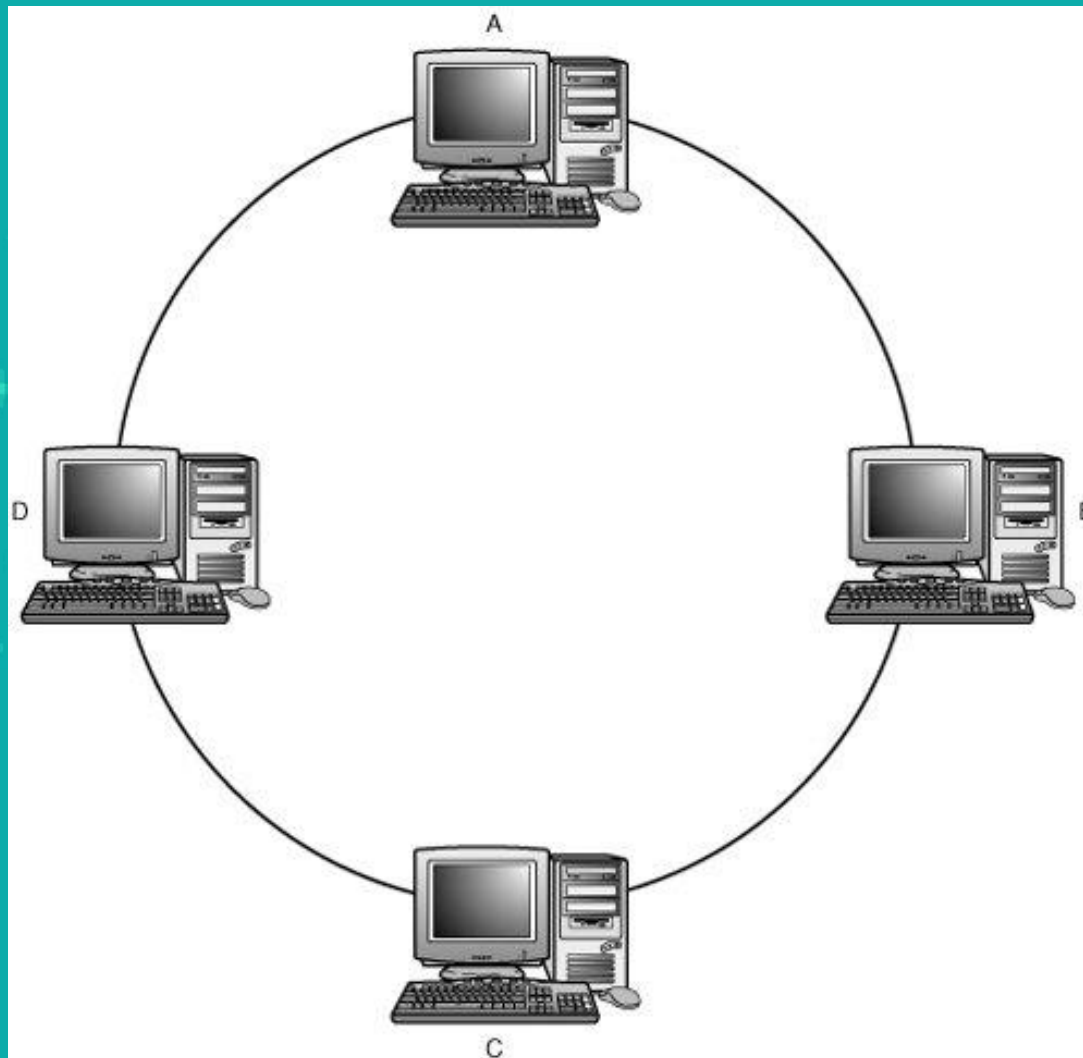
Advantages

- Easy to add devices as the network expands
- One cable failure does not bring down the entire network (resilience)
- Hub or switch provides centralized management
- Easy to find device and cable problems
- Can be upgraded to faster speeds
- Lots of support as it is the most used

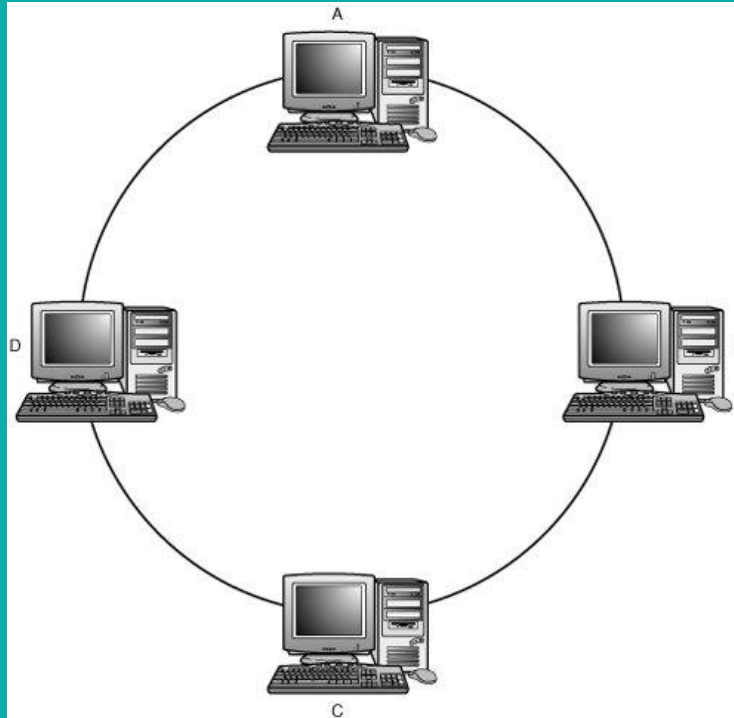
Disadvantages

- Failure of the central hub can bring down the entire network
- A star network requires more cable than a ring or bus network
- Costs are higher (installation and equipment) than for most bus networks

Ring Topology



Ring Topology



Similar to a bus network, rings have nodes daisy chained, but the end of the network in a ring topology comes back around to the first node creating a complete circle.

Ring Topology

- Each node takes a turn sending and receiving information through the use of Token.
- The token along with any data is sent from the first node to the second node which extracts the data addresses to it and adds any data it wishes to send. Then second node passes the token and data to the third node, etc until it comes back around to the first node again.
- So only the node with the token is allowed to send data.

Ring Topology

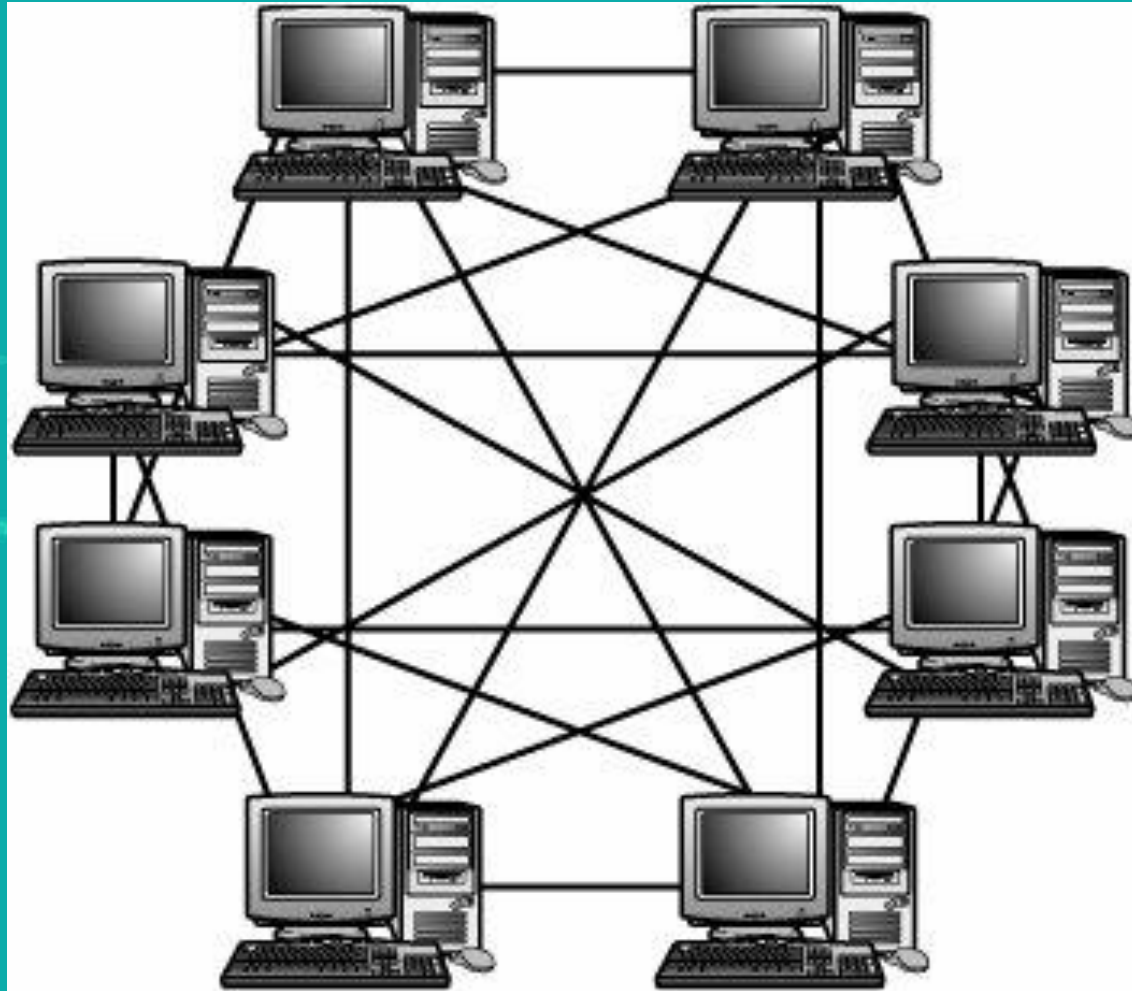
Advantages

- Data packets travel at great speed
- No collisions
- Easier to fault find
- No terminators required

Disadvantages

- Requires more cable than a bus
- A break in the ring will bring it down
- Not as common as the bus – less devices available

Mesh Topology



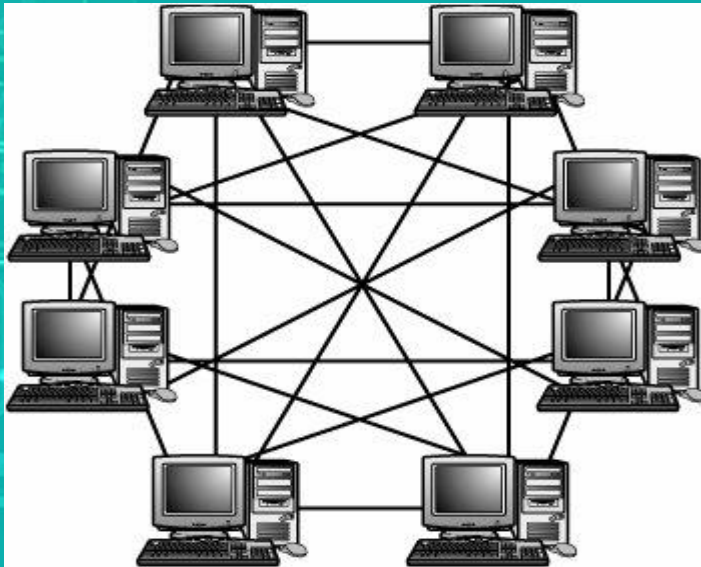
Mesh Topology

- Each node is connected to several other nodes
- Allows communication to continue in the event of a breakdown in any one connection
- It is “Fault Tolerant”
- There are two types of mesh:
 - Full
 - Partial

Mesh Topology

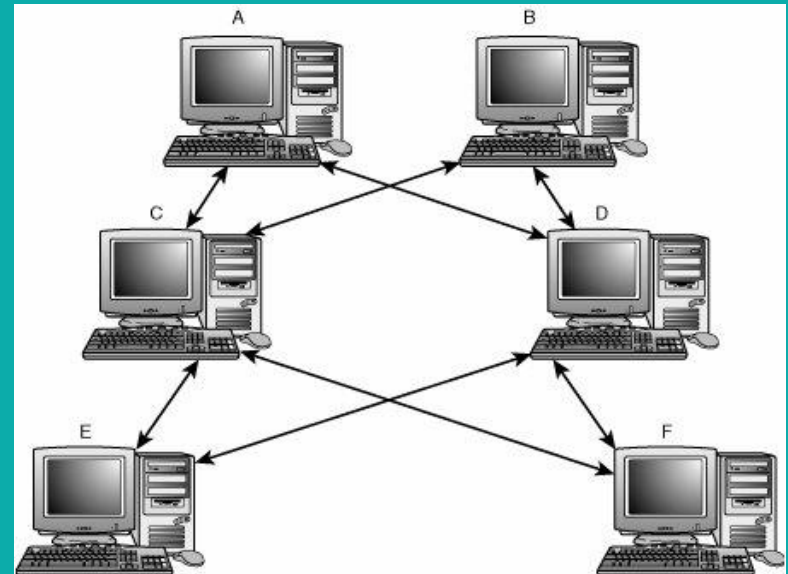
Full Mesh Topology

Full mesh means that every node contained in a network has a connection to every other node contained in the network.



Partial Mesh Topology

Partial meshes are designed to provide redundancy where it is needed.



Mesh Topology

Advantages

- Improves Fault Tolerance

Disadvantages

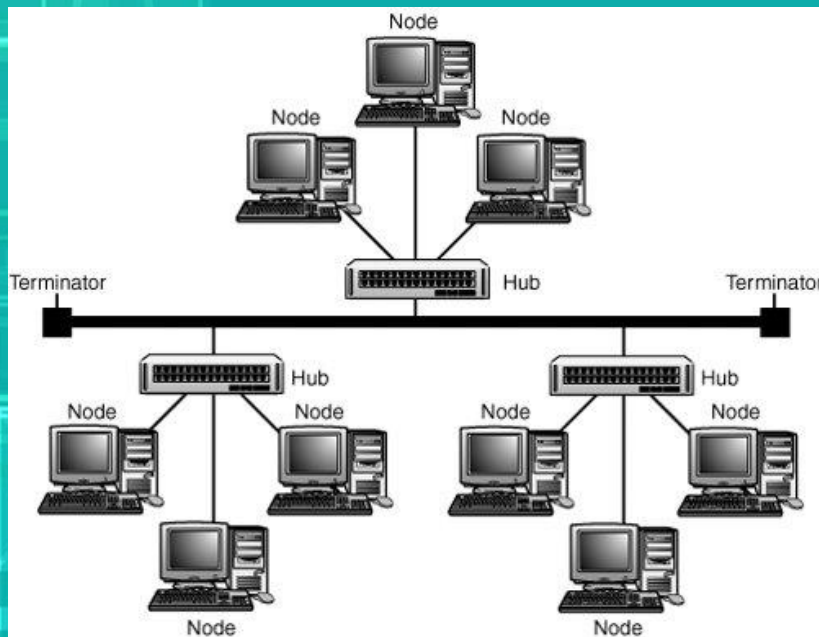
- Expensive
- Difficult to install
- Difficult to manage
- Difficult to troubleshoot

Hybrid Topology

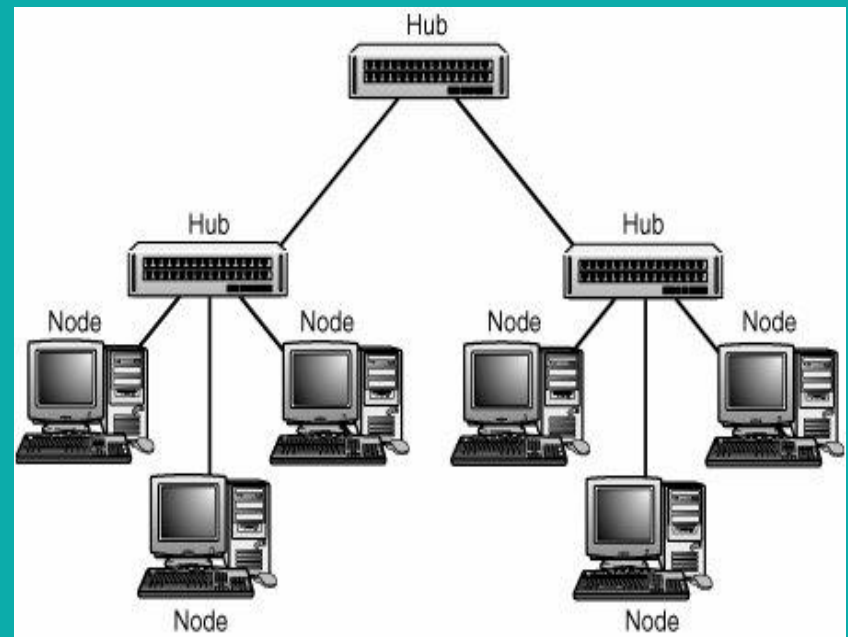
- Hybrid networks combine two or more topologies in such a way that the resulting network does not exhibit one of the standard topologies.
- A hybrid topology is always produced when two different basic network topologies are connected.
- Major types:
 - Tree network (or star-bus network)
 - Hierarchical Star

Hybrid Topology

Tree network



Hierarchical Star



Standards

- Standards are set of rules that ensure hardware and software released from different companies work together.
- Example of organizations that coordinate standards:
 - **International Organization for Standardization (ISO)** - Federation of standards organizations from multiple nations.
 - **American National Standards Institute (ANSI)** - Responsible for coordinating and publishing computer and information technology standards in the United States.
 - **International Electrical and Electronics Engineers (IEEE)** - Professional organization for the electrical and electronics field.